

The M-CATs

SEMINAR 15

Table of Contents

CH 6

Organic Chemistry

A&K 1: Electrophiles & Redox

CH 7

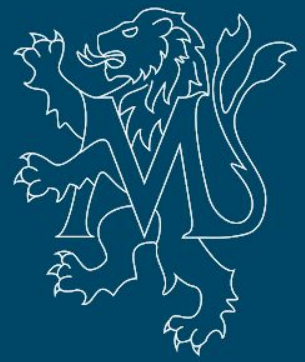
Organic Chemistry

A&K 2: Enolates

CH 8

Organic Chemistry

Carboxylic Acids

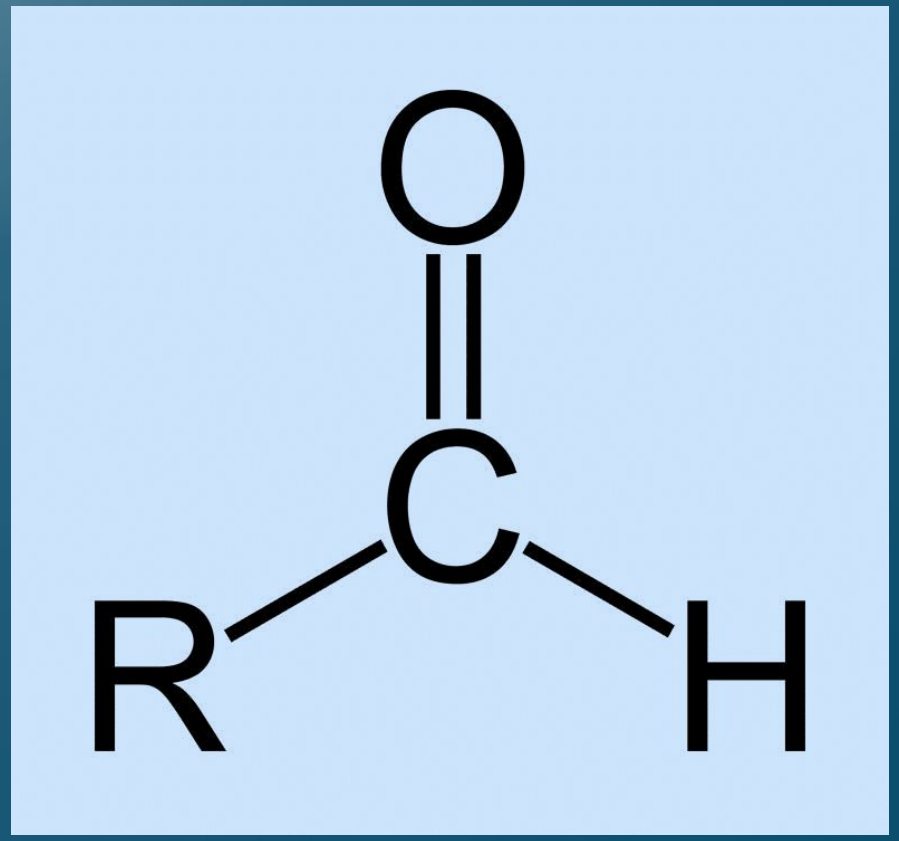


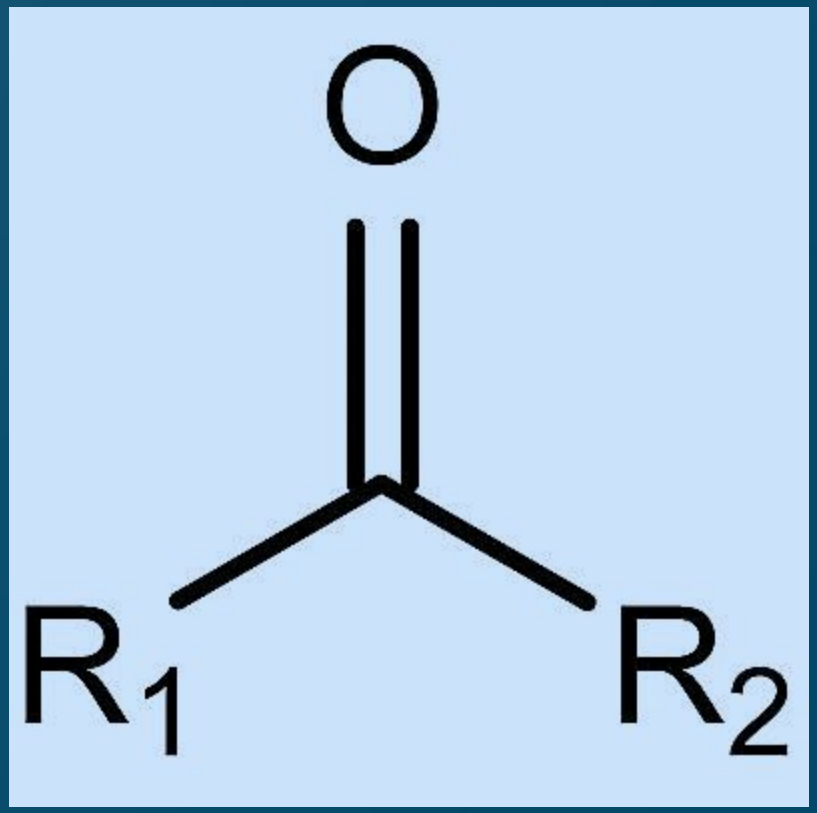
Organic Chemistry Chapters 6, 7

Aldehydes & Ketones

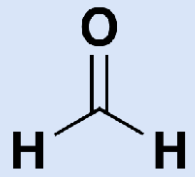


6.1. Description & Properties

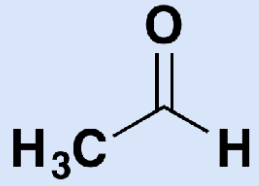




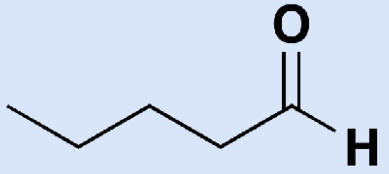
Nomenclature: Aldehydes



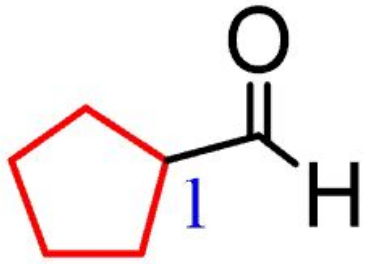
methanal
(formaldehyde)



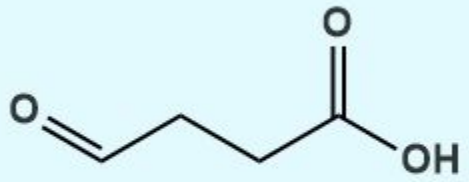
ethanal
(acetaldehyde)



pentanal
(valeraldehyde)

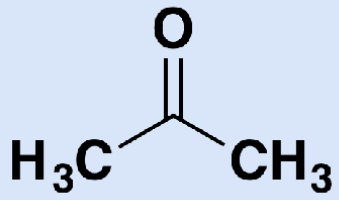


cyclopentanecarbaldehyde

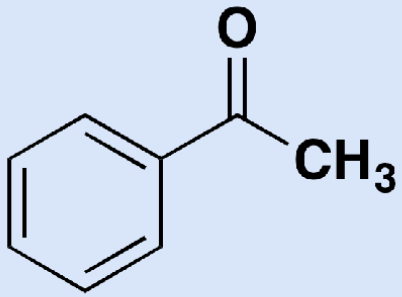


4-oxobutanoic acid

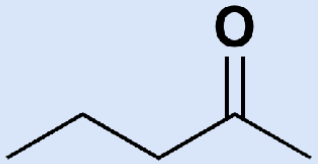
Nomenclature: Ketones



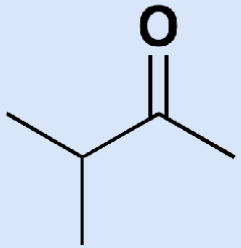
propanone
(acetone)



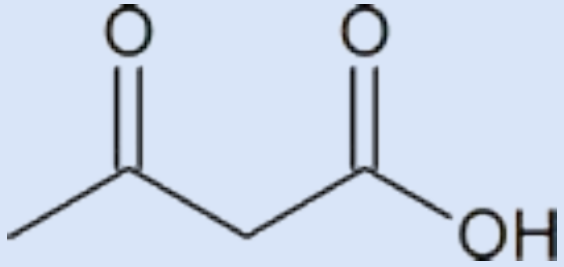
acetophenone
(methyl phenyl ketone)



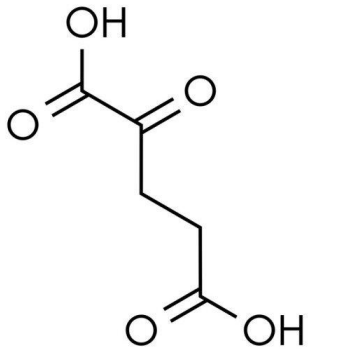
2-pentanone
(methyl propyl ketone)



3-methyl-2-butanone
(methyl isopropyl ketone)

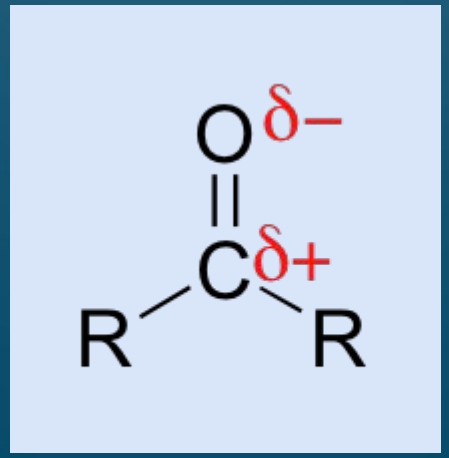


3-oxobutanoic acid



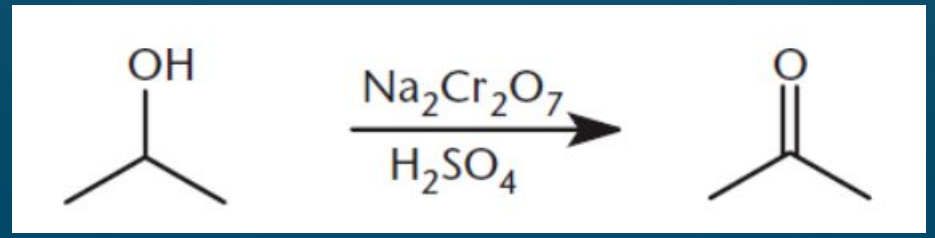
alpha-ketoglutaric acid

Physical Properties



Formation

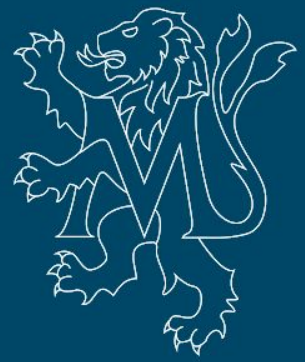




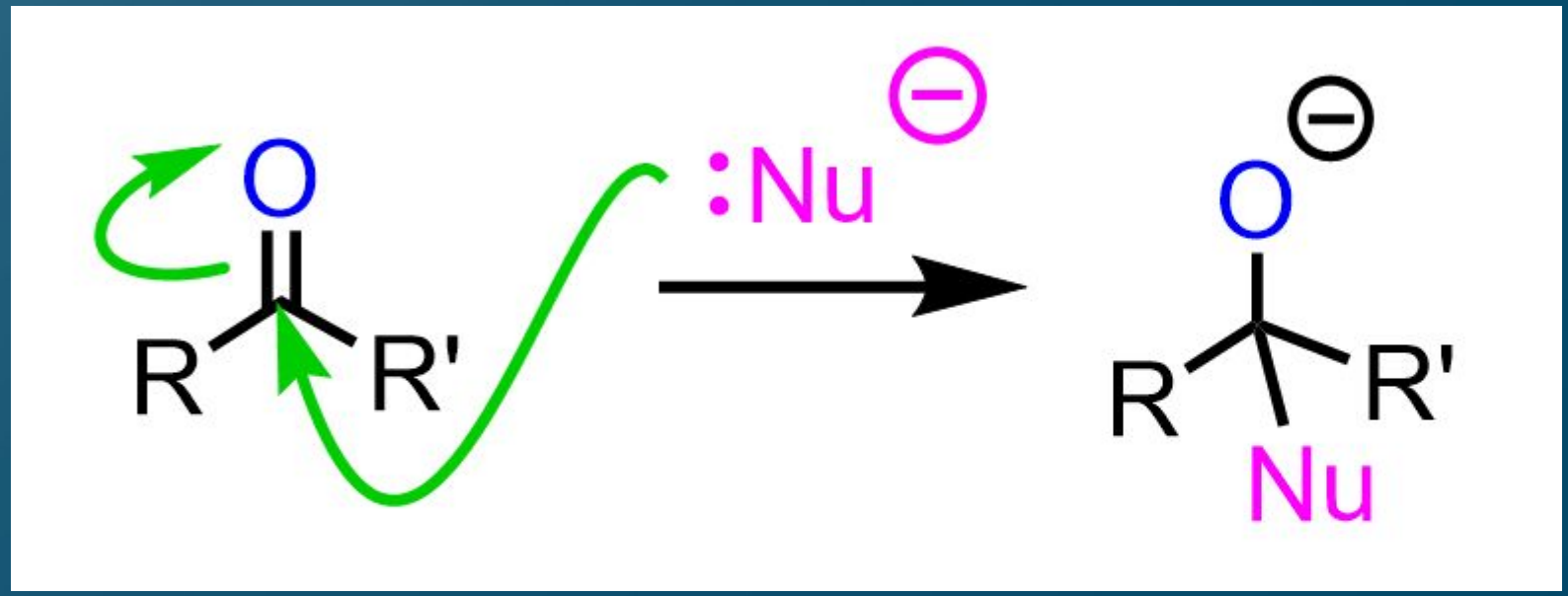
Question

Can you directly reduce carboxylic acids into ketones and/or aldehydes?

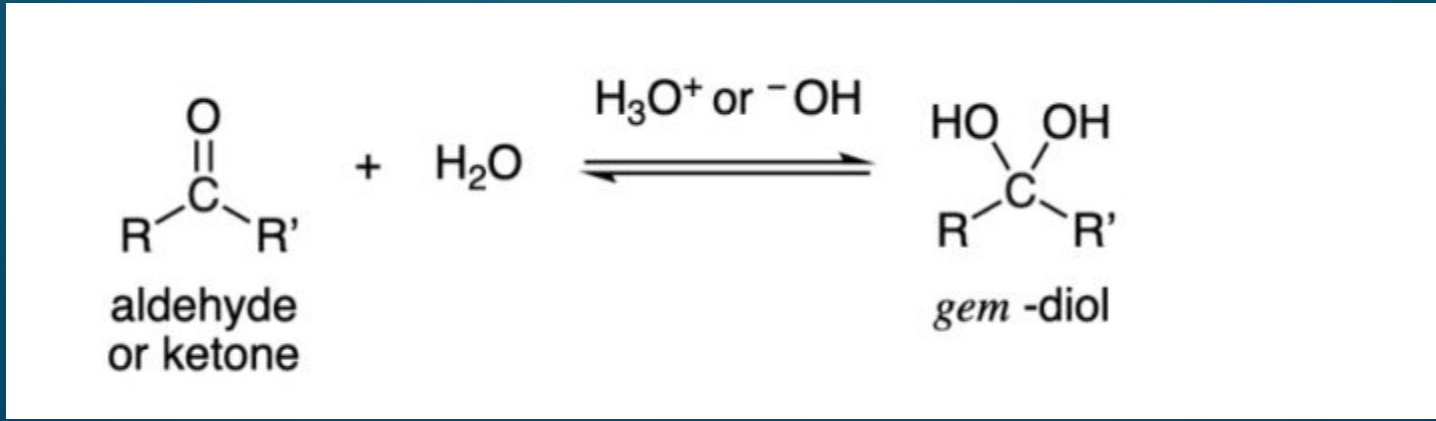
A. No—they would reduce all the way to alcohols
B. No—you would end up with unwanted byproducts
C. Yes—works as the reverse of reduction
D. Yes—but only ketones b/c of increased substitution



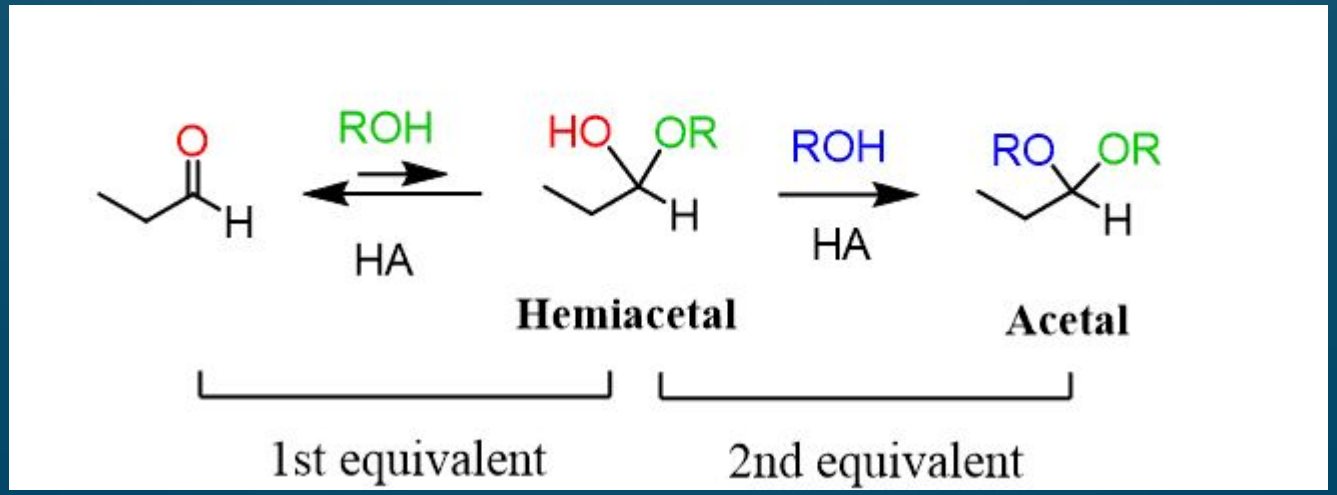
6.2. Nucleophilic Addition Reactions

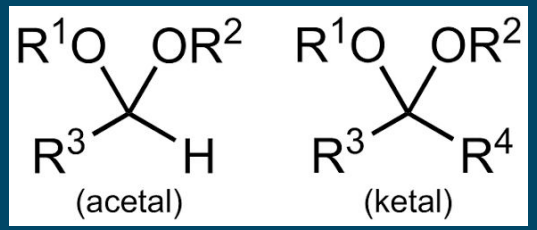


Hydration

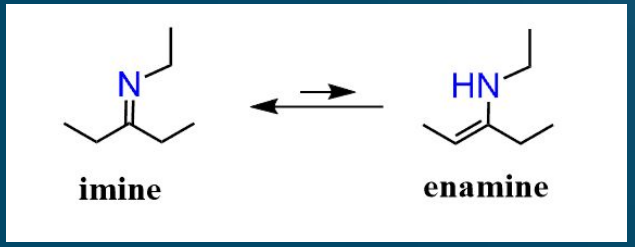


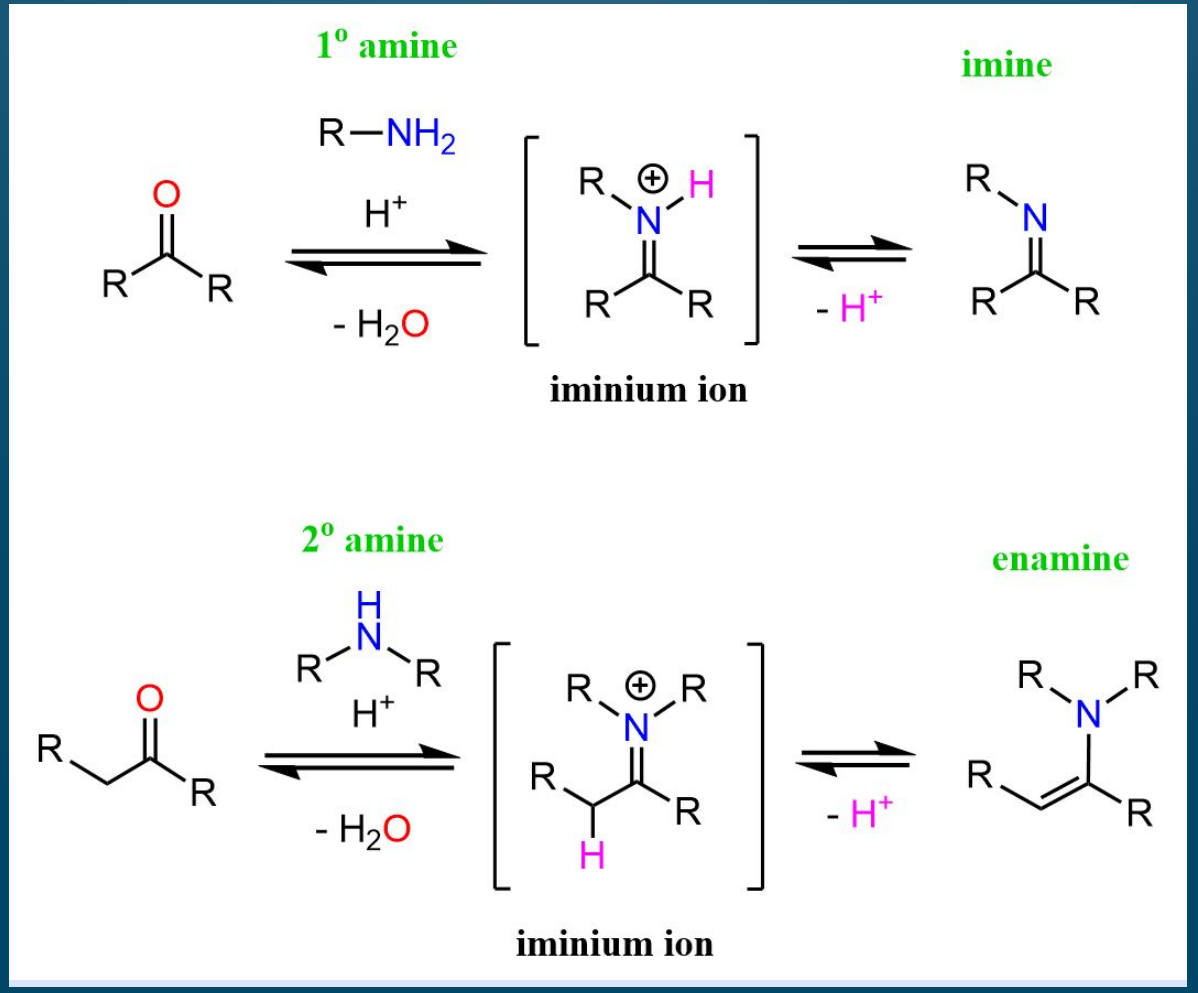
Acetal formation



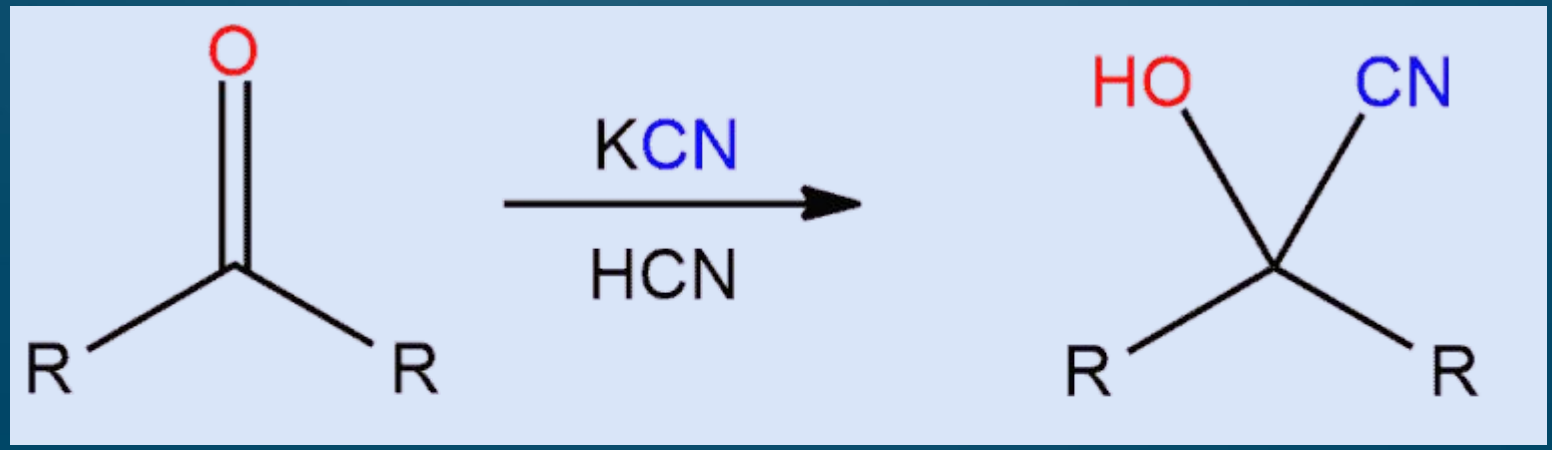


Imines and Enamines

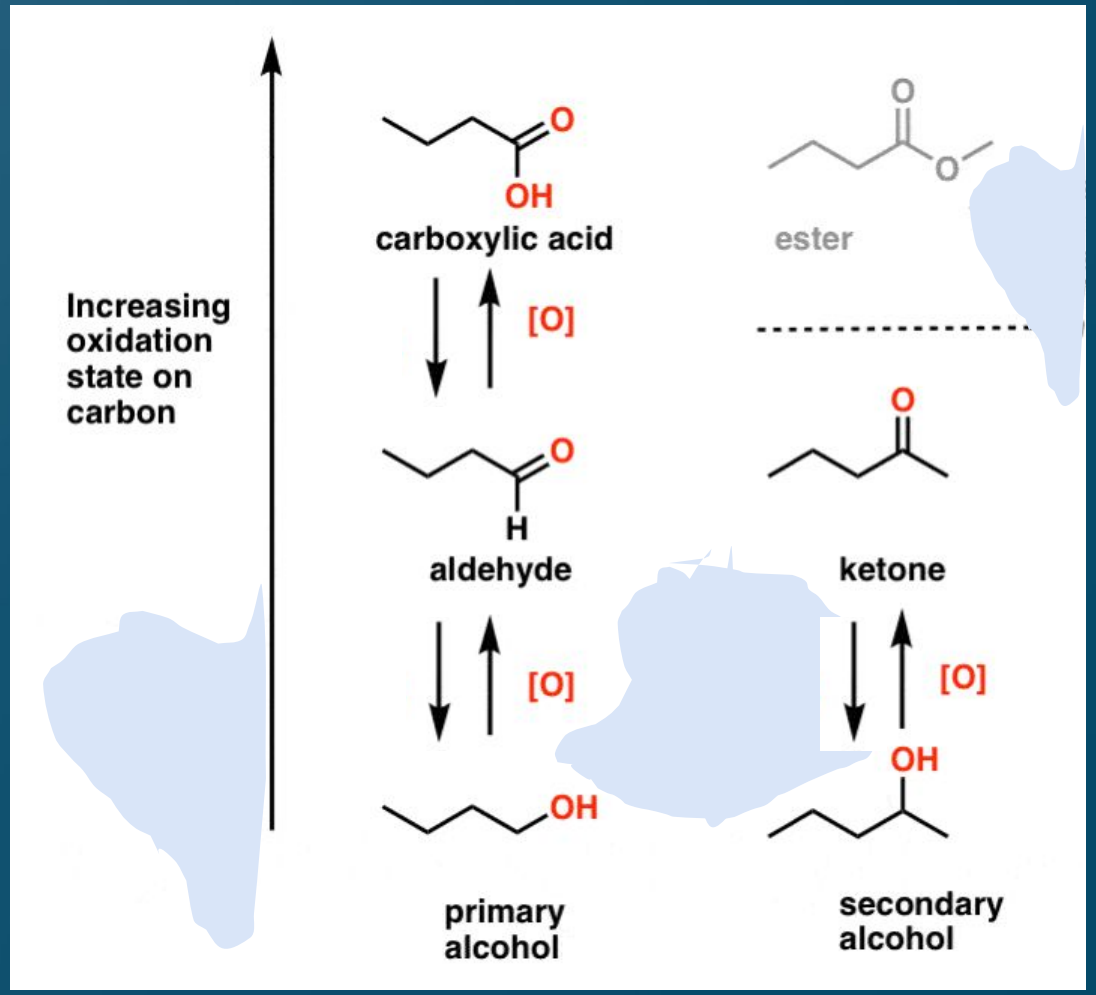




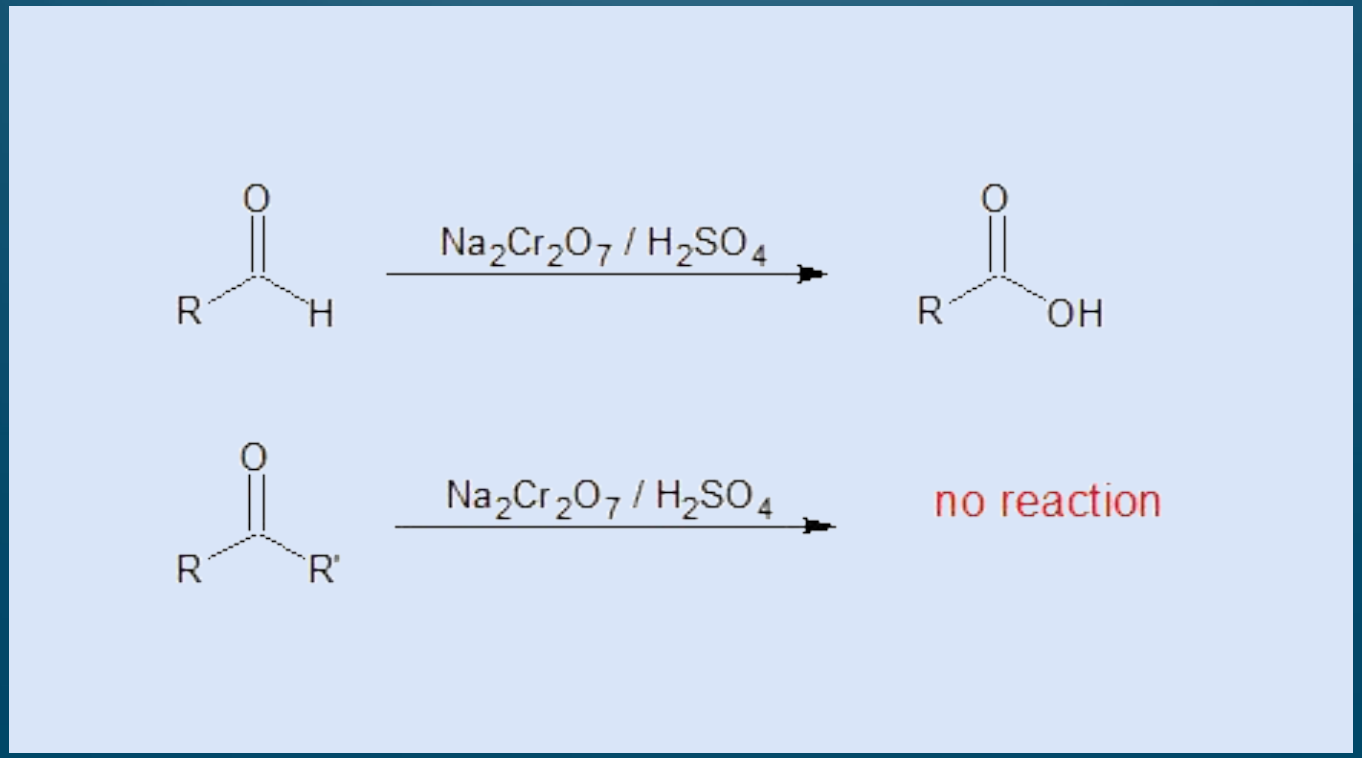
Cyanohydrins



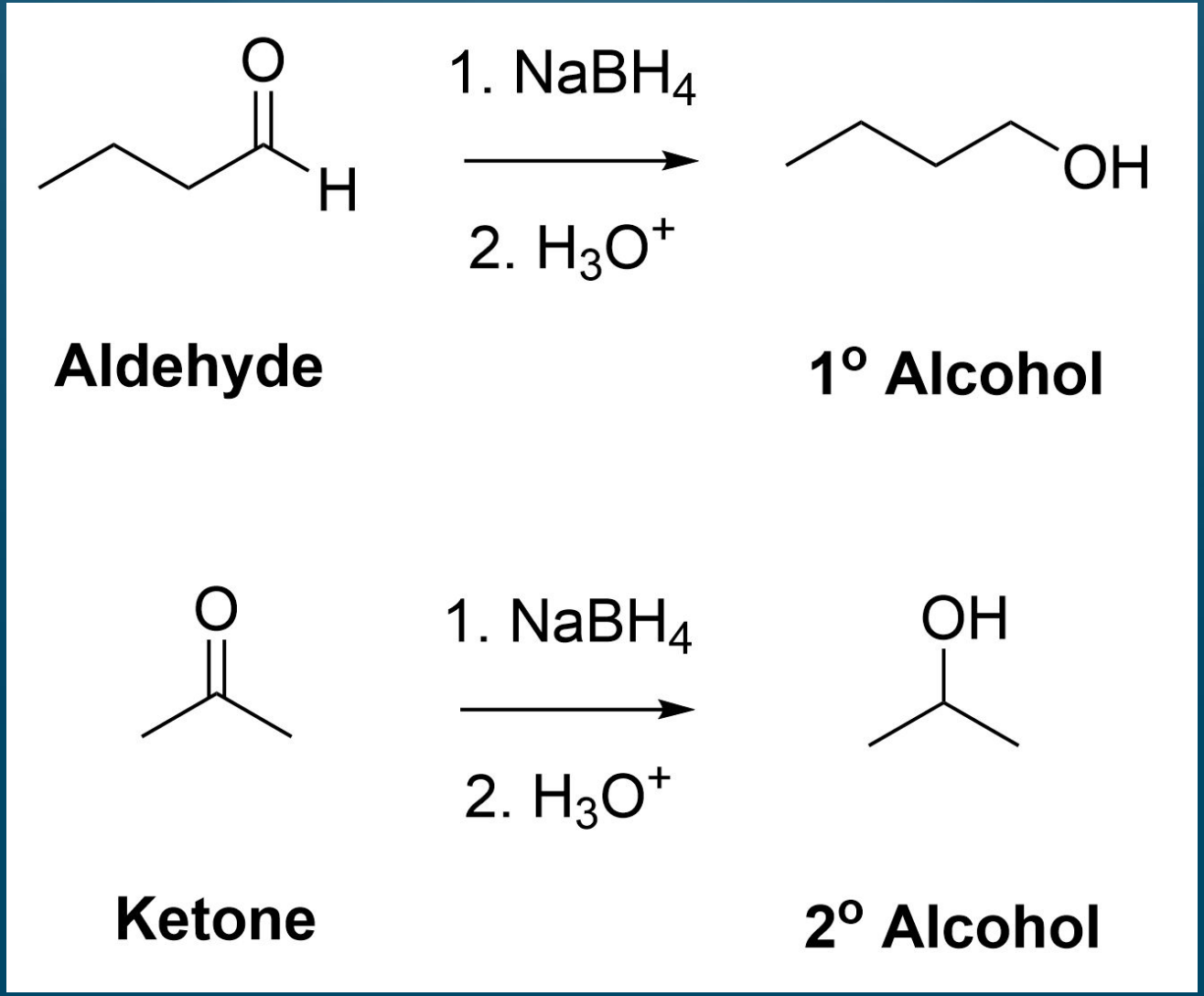
6.3. Carbonyl Redox Reactions



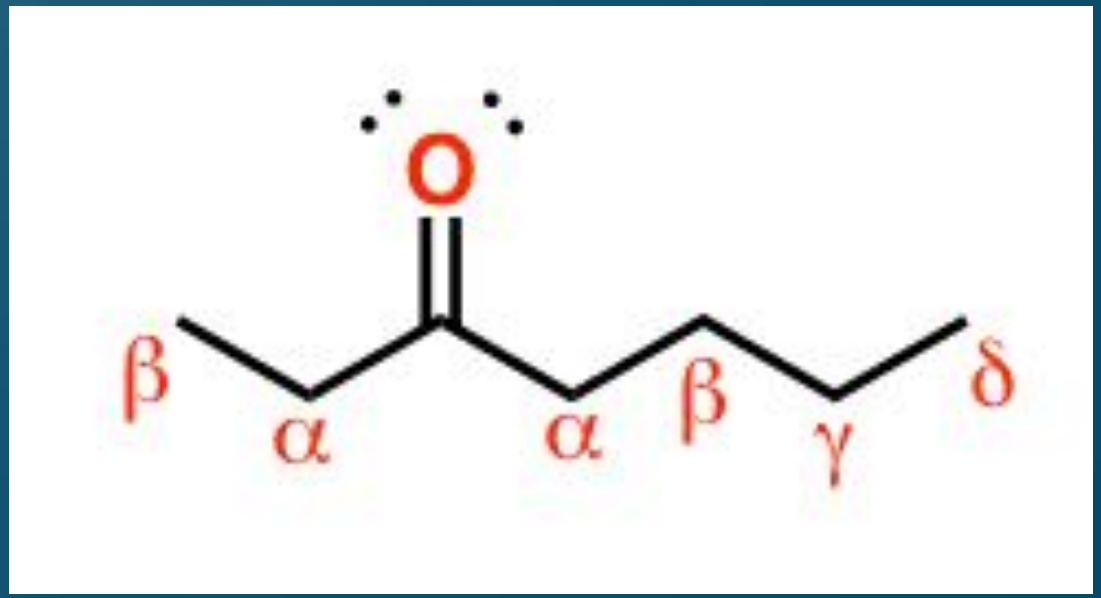
Oxidation



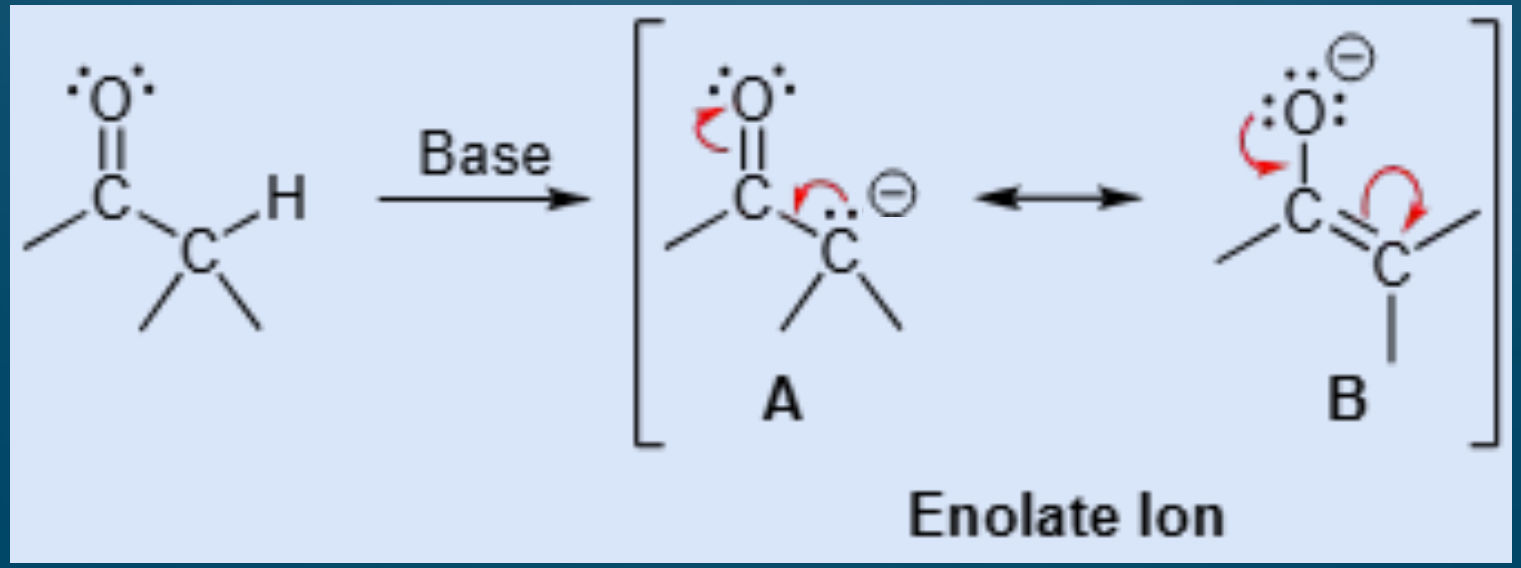
Reduction


O
H
1. NaBH4
2. H3O+
OH
Aldehyde
1° Alcohol
O
1. NaBH4
2. H3O+
OH
Ketone
2° Alcohol

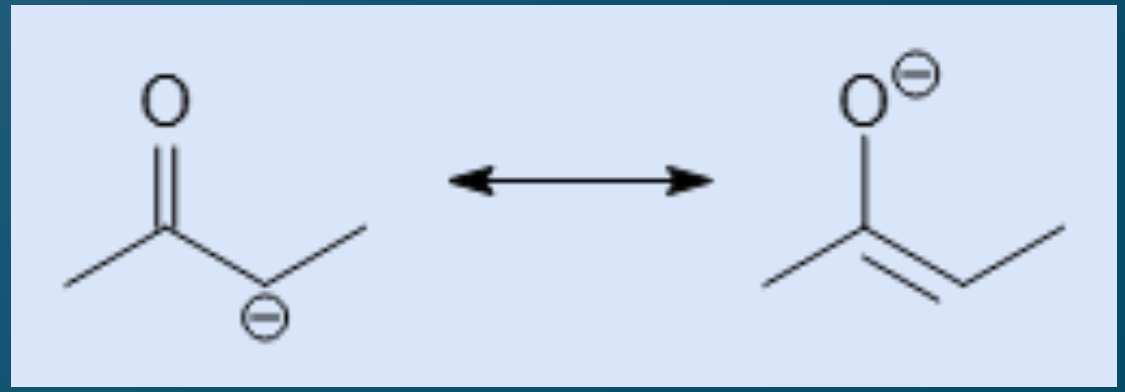
7.1. Principles of α -Hydrogens

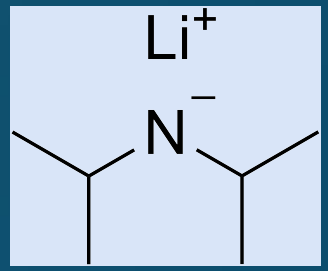


Acidity

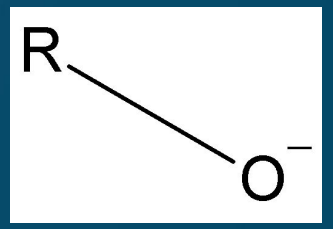


7.2. Enolate Chemistry

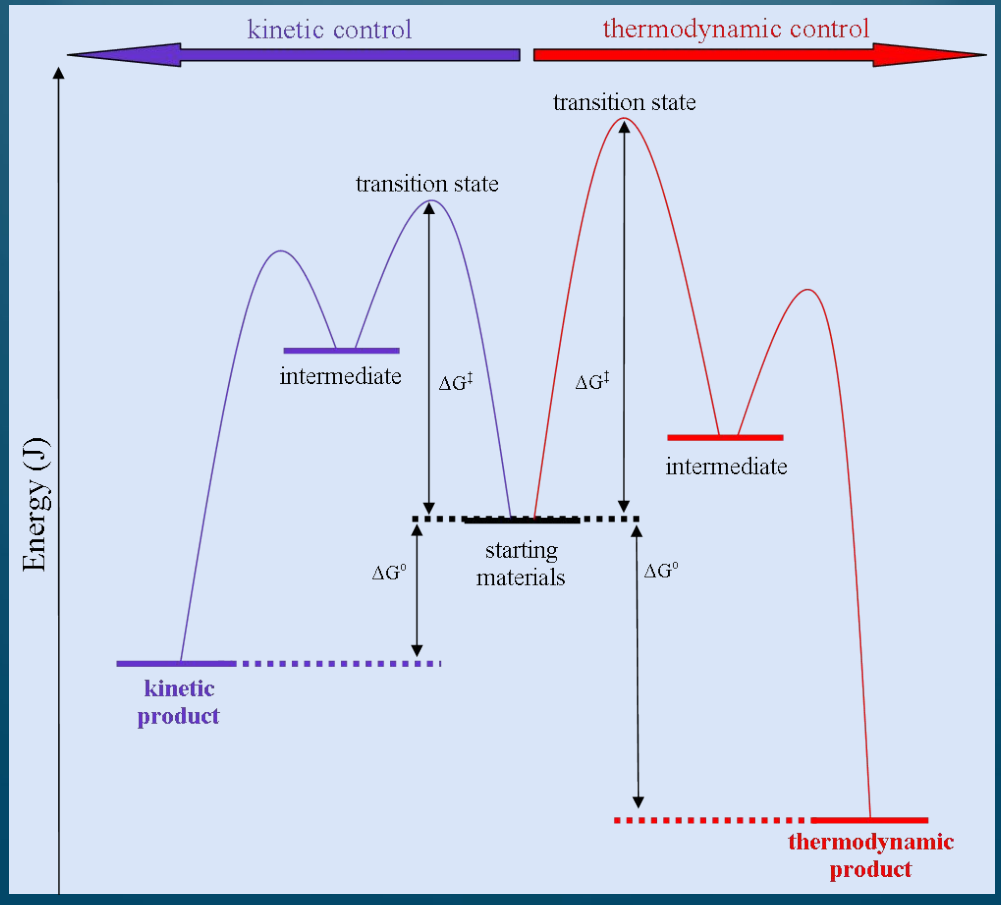


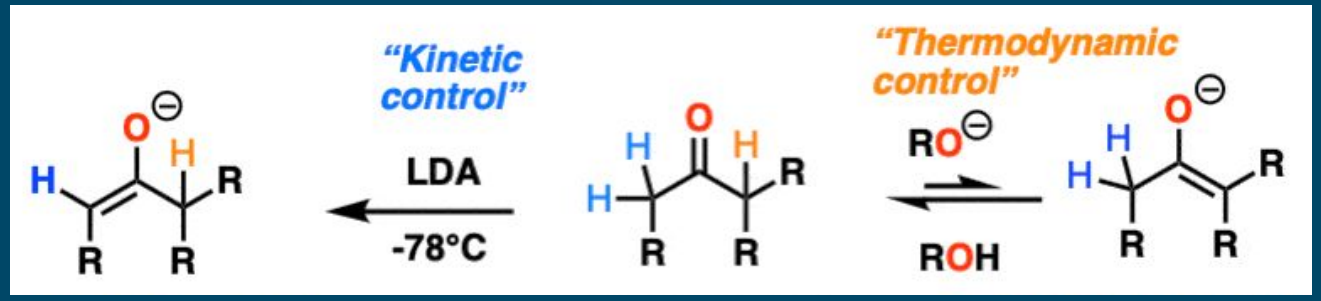




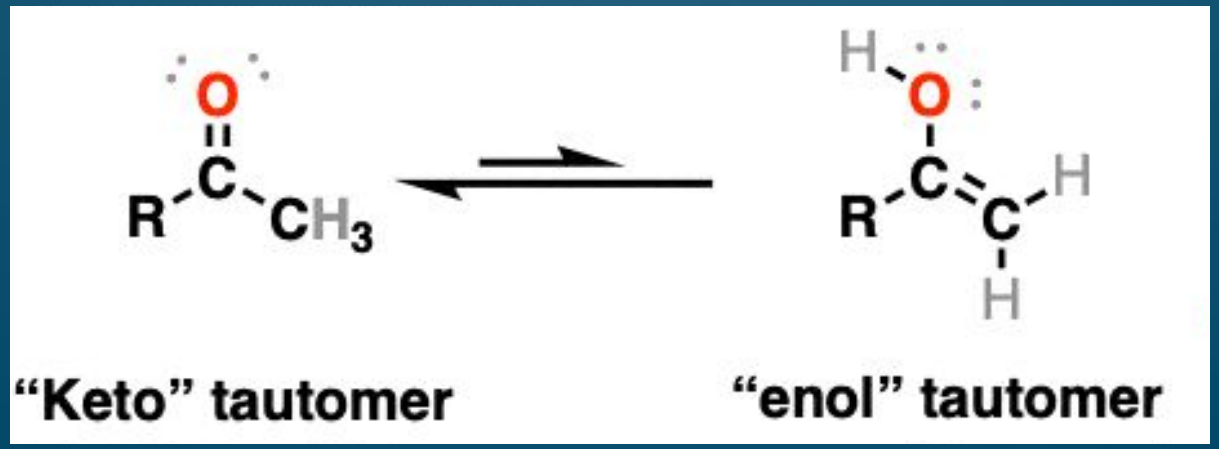


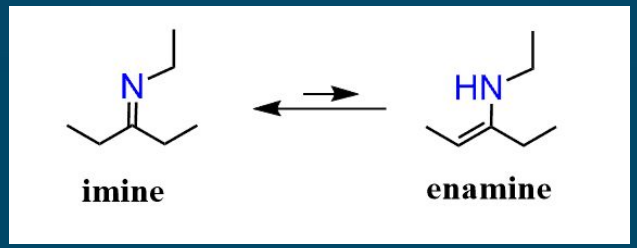
Thermodynamic/Kinetic Control





Tautomerization





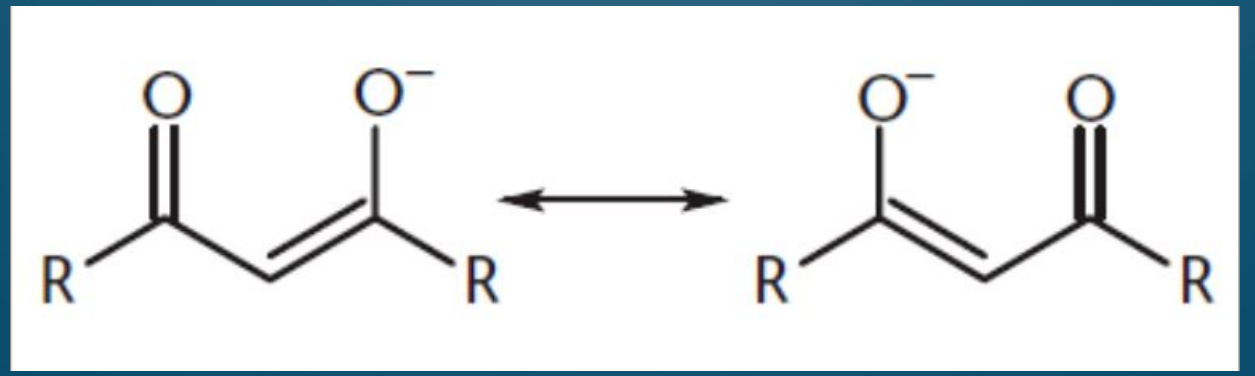
Single Bond Energies (kJ/mol of bonds)

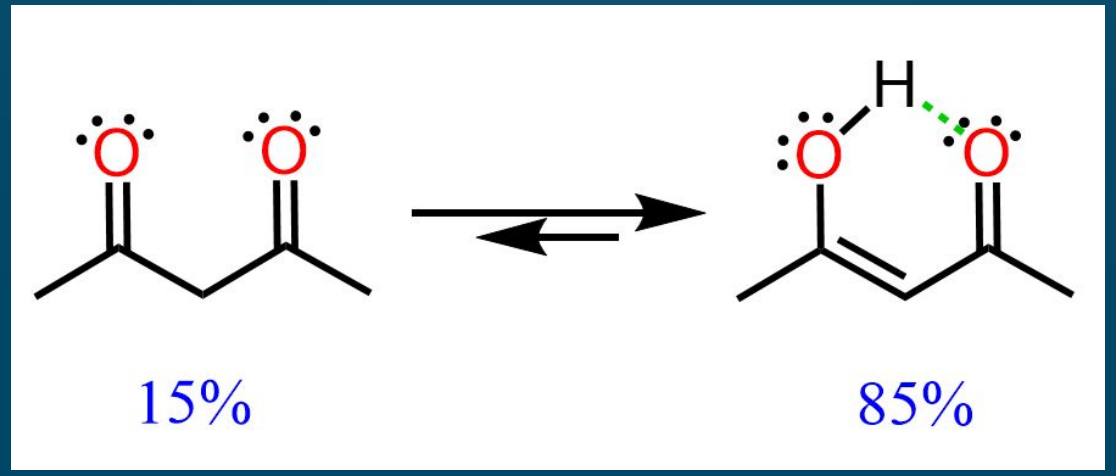
	H	C	N	O	S	F	Cl	Br	I
H	436								
C	413	346							
N	391	305	163						
O	463	358	201	146					
S	347	272	—	—	226				
F	565	485	283	190	284	155			
Cl	432	339	192	218	255	253	242		
Br	366	285	—	201	217	249	216	193	
I	299	213	—	201	—	278	208	175	151

Multiple Bond Energies (kJ/mol of bonds)

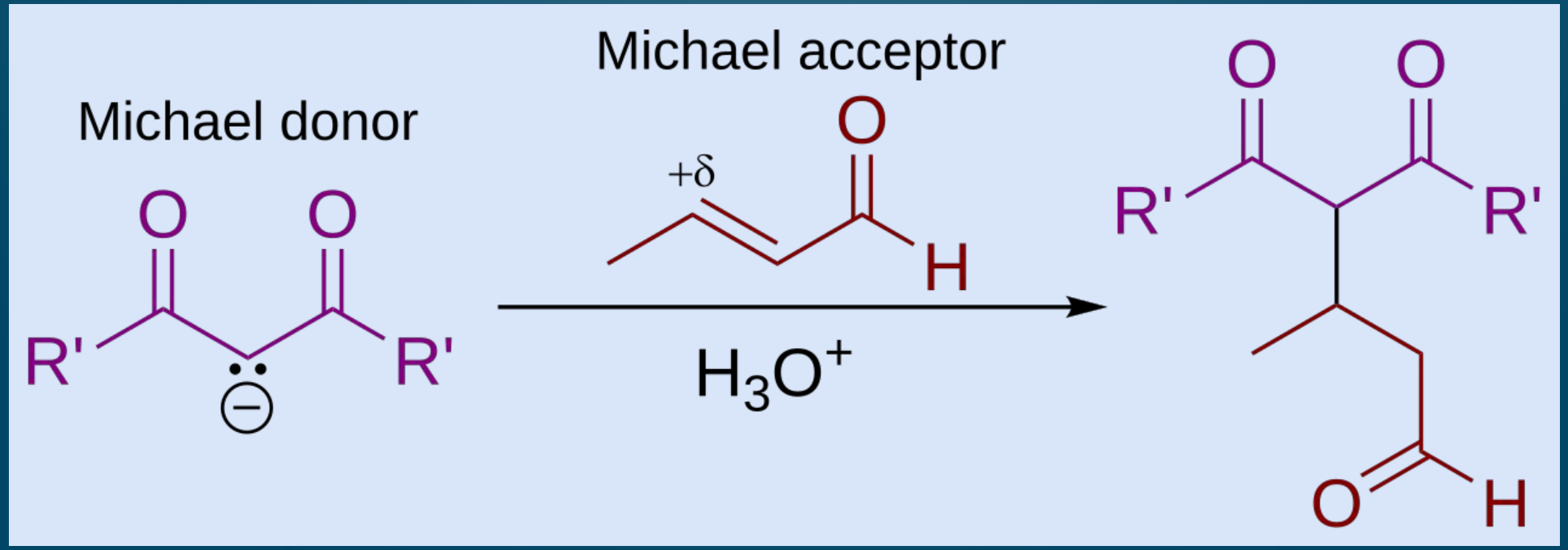
C=C 602	C=N 615	C=O 799
C≡C 835	C≡N 887	C≡O 1072
N=N 418	N=O 607	
N≡N 945	O=O 498	

β-Diketones

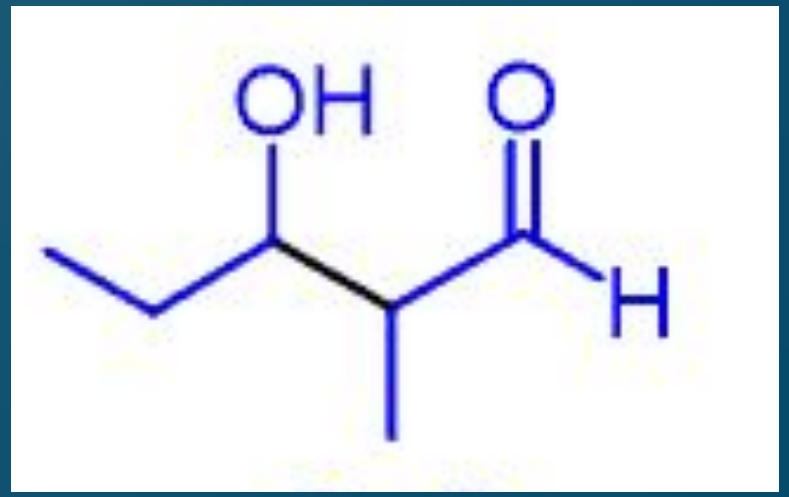




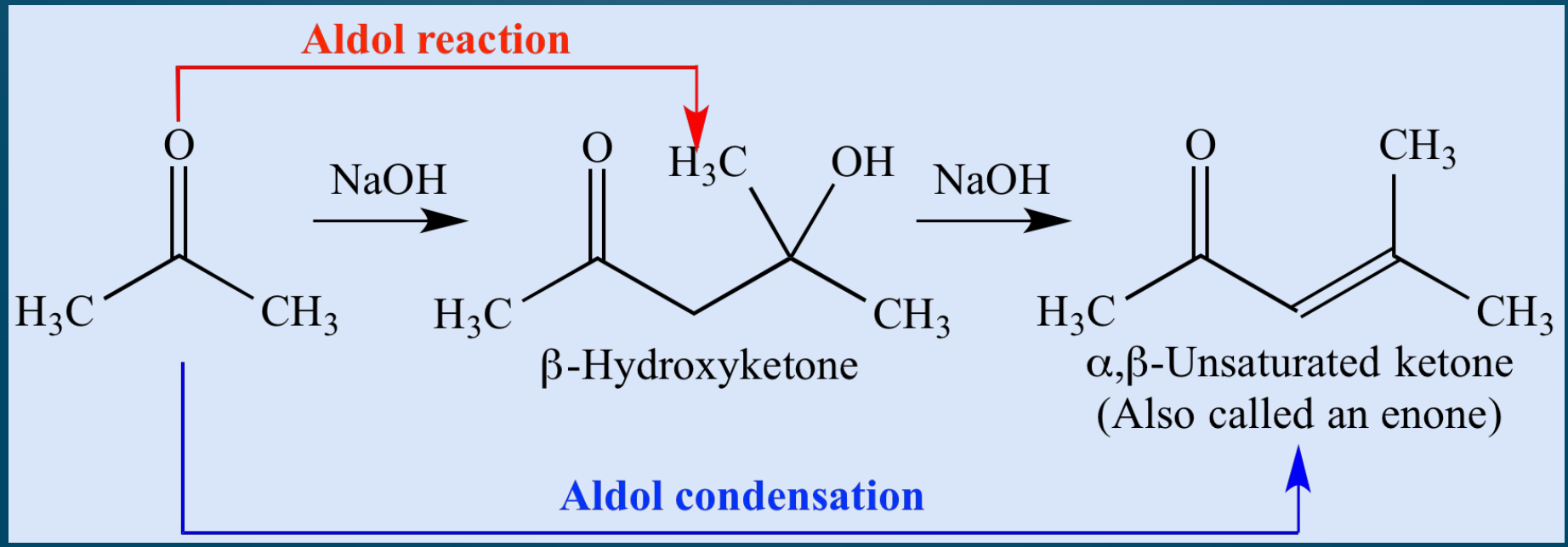
Michael Addition



7.3. Aldols



Aldol Condensation

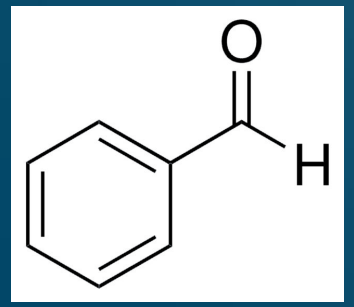


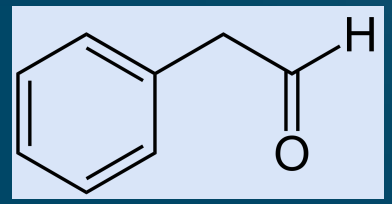
Retro Aldol Reaction

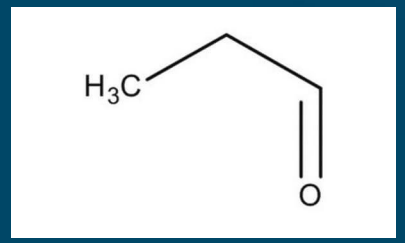


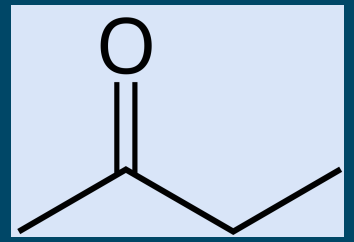
Question

Which of these compounds will not undergo an aldol reaction?

A. 

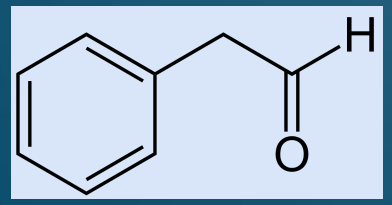
B. 

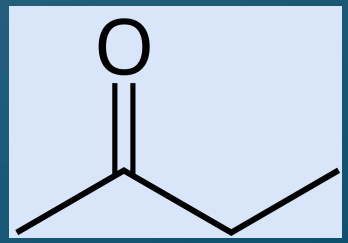
C. 

D. 

Question

The following two aldols (pictured)are combined in an aldol reaction at a high temperature. How many different aldols may be produced?





A. 0

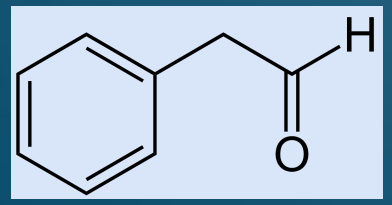
B. 1

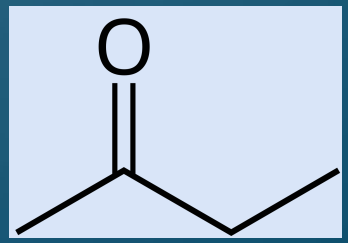
C. 2

D. 4

Question

The following two aldols (pictured) are combined in an aldol reaction at a *low* temperature. How many different aldols may be produced?





A. 0

B. 1

C. 2

D. 4